

Exact window products, sharp endpoint optimization, and original relation costs

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This continuation proves the product refinement in the delivered four-endpoint calculation, then solves the associated scalar problem with only the combined logarithmic budget retained. It gives explicit maps from these quantities to the original source, relation and Schur determinants. The new arithmetic consequence uses the proved norm estimate of the accompanying *Arithmetic endpoint bounds* source. Its analytic proof remains an explicit dependency; the finite polynomial calculations below do not certify that analytic theorem or an upper estimate for the growing arithmetic relation volumes.

1 Original source and a proof of its exact radius identity

Fix the original monic packet h of actual selected zeros, with all their orders, of $g = 2\xi$, and put

$$v_h = g/h, \quad w_h(t) = |v_h(1/2 + it)|^2/(2\pi), \quad m_{h,k} = w_h^{*k}, \quad c = k/2, \quad \int m_{h,k} = \mu_h^k, \quad \mu_h = \int w_h. \quad (\text{EP.1})$$

The complete source construction proves positivity almost everywhere and all polynomial moments, also after multiplication by $e^{\theta u}$ for real $|\theta| < \pi/2$. Retain that literal density and the pairing

$$\langle P, Q \rangle_\theta = \int_{\mathbb{R}} \overline{P(c + iu)} Q(c + iu) e^{\theta u} m_{h,k}(u) du. \quad (\text{EP.2})$$

It is conjugate-linear in its first entry. Every nonzero polynomial has positive squared norm since its zero set on the line is finite. Let χ be the original degree- q monic generator of $(h(s_1), \dots, h(s_k)) \cap \mathbb{C}[S]$, where $S = \sum s_i$. In this section $q \geq 1$ and the packet is stable under $\rho \mapsto 1 - \bar{\rho}$. Reflection preserves this ideal and gives $\bar{\chi}(k - S) = (-1)^q \chi(S)$ by monicity. Hence $\hat{\chi}(u) = i^{-q} \chi(c + iu)$ is real monic.

Write $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$, $\mathcal{P}_{-1} = 0$, and $C = \mathbb{C}[S]/(\chi)$ in the coordinates $1, S, \dots, S^{q-1}$. Monic division proves the exact sequence

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{\times \chi} \mathcal{P}_N \xrightarrow{J_N} C \longrightarrow 0 \quad (N \geq q-1). \quad (\text{EP.3})$$

Let H_N be the Gram in the stated monomial coordinates. The exact least-lift matrices, with their full types, are

$$K_N = J_N H_N^{-1} J_N^*, \quad G_N = K_N^{-1}, \quad R_N = H_N^{-1} J_N^* G_N : C \longrightarrow \mathcal{P}_N, \quad V_N = \det G_N. \quad (\text{EP.4})$$

Surjectivity of J_N proves $K_N > 0$. If B_N is the multiplication matrix for χ , direct multiplication gives $J_N R_N = I$ and $B_N^* H_N R_N = 0$. Every lift of x is $R_N x + B_N y$ and has squared norm $x^* G_N x + y^* B_N^* H_N B_N y$. This proves both the minimum property and $R_N^* H_N R_N = G_N$. Inclusions of the polynomial spaces therefore give $G_{N+1} \preceq G_N$ and $V_{N+1} \leq V_N$.

Let p_j be the monic orthogonal family in S , with squared norms $\omega_j > 0$, and let $b_j = [p_j]_\chi$. The monic triangular coefficient matrix conjugates H_N to $\text{diag}(\omega_0, \dots, \omega_N)$, so

$$K_N = \sum_{j=0}^N b_j b_j^* / \omega_j. \quad (\text{EP.5})$$

For $N \geq q$ define $\delta_N = V_N / V_{N-1} \in (0, 1]$. The determinant identity $\det(I + XY) = \det(I + YX)$ follows by eliminating the two diagonal blocks of $\begin{pmatrix} I & X \\ -Y & I \end{pmatrix}$ in the two orders. Applied to (EP.5), it gives

$$\frac{b_N^* G_N b_N}{\omega_N} = 1 - \delta_N, \quad \frac{b_{N+1}^* G_N b_{N+1}}{\omega_{N+1}} = \delta_{N+1}^{-1} - 1. \quad (\text{EP.6})$$

Both follow respectively by subtracting the last term of K_N and adding the next term; all the earlier kernels inverted here are positive.

The exact algebra map

$$\Phi : C \longrightarrow \mathbb{C}[u]/(\hat{\chi}), \quad [P] \longmapsto [P(c + iu)], \quad \Phi^{-1}[Q] = [Q((S - c)/i)] \quad (\text{EP.7})$$

has coefficient matrix E triangular with diagonal $1, i, \dots, i^{q-1}$. Thus $|\det E| = 1$. The quotient Gram becomes $E^{-*} G_N E^{-1}$ and its determinant stays V_N . Put $A = M_S : C \rightarrow C$ and $T = M_u$ in the real quotient, so $\Phi A \Phi^{-1} = cI + iT$ and $\sigma = \text{Tr } T = (\text{Tr } A - cq)/i \in \mathbb{R}$. The real monic polynomials $Q_j = i^{-j} p_j(c + iu)$ have norms ω_j . Pairing uQ_j with Q_l proves

$$uQ_j = Q_{j+1} + a_j^{(0)} Q_j + (\omega_j / \omega_{j-1}) Q_{j-1}, \quad a_j^{(0)} \in \mathbb{R}; \quad Sp_j = p_{j+1} + (c + ia_j^{(0)}) p_j - (\omega_j / \omega_{j-1}) p_{j-1}. \quad (\text{EP.8})$$

The last term is absent for $j = 0$. Indeed pairing with degrees below $j - 1$ vanishes by self-adjoint multiplication by u , the leading coefficient of Q_{j+1} is one, and pairing with Q_{j-1} gives the displayed norm ratio.

For completeness the phase has its original derivative interpretation. In the real coordinates let $d_j = [Q_j]$, and keep the transported matrices denoted temporarily by R, G . Differentiating the minimum norm shows $G' = R^* u R$: the two terms containing R' vanish because $J_N R' = 0$ and R is orthogonal to $\ker J_N$. The leading coefficient of Rx is $d_N^* Gx / \omega_N$. The polynomial

$$uRx - RTx - (Q_{N+1} - Rd_{N+1})d_N^* Gx / \omega_N$$

is in $\ker J_N$: its leading degree cancels and its reduction is zero. Pair it with R and use $R^* Q_{N+1} = 0$. Taking the trace gives

$$\partial_\theta \log V_N = \sigma - d_N^* G d_{N+1} / \omega_N, \quad \phi_N := \sigma - \partial_\theta \log V_N, \quad b_N^* G_N b_{N+1} = i\omega_N \phi_N. \quad (\text{EP.9})$$

The factor i is $i^{\overline{N}} i^{N+1}$; no phase has been removed. Differentiation is justified by the exponential moment domination of the original source.

Substitute (EP.8) into (EP.5). Terms with adjacent indices cancel because ω_j / ω_{j-1} is the exact recurrence coefficient; diagonal terms cancel against $2cK_N$. Thus

$$AK_N + K_N A^* - 2cK_N = (b_{N+1} b_N^* + b_N b_{N+1}^*) / \omega_N. \quad (\text{EP.10})$$

The relative Hermitian defect $\mathcal{H}_N = G_N^{-1/2} (A^* G_N + G_N A - 2cG_N) G_N^{-1/2}$ has rank at most two and trace zero. For $x = G_N^{1/2} b_N$, $y = G_N^{1/2} b_{N+1}$ its matrix is $(xy^* + yx^*) / \omega_N$. Expanding its square gives trace $2(\|x\|^2 \|y\|^2 - |x^* y|^2) / \omega_N^2$ because $x^* y = i\omega_N \phi_N$. Its nonzero eigenvalues, if present, are $\epsilon_N, -\epsilon_N$ with $\epsilon_N > 0$; at rank zero define $\epsilon_N = 0$. Consequently (EP.6) proves the full identity

$$\boxed{\epsilon_N^2 + \phi_N^2 = \frac{\omega_{N+1}}{\omega_N} (1 - \delta_N) (\delta_{N+1}^{-1} - 1).} \quad (\text{EP.11})$$

For $q = 1$ the trace-zero defect is the zero map; the identity retains its possibly nonzero phase. We never form V_{q-2} .

2 Window product, endpoint extremum, and phases

Fix $n \geq q$, $r \geq 1$, $m = n + r$, and put

$$d_j = \delta_{n+j} \ (0 \leq j \leq r), \quad U = \omega_m / \omega_n, \quad \mathcal{A} = U^{1/(2r)}, \quad E_* = \min_{0 \leq j < r} \epsilon_{n+j}. \quad (\text{EP.12})$$

Multiplying every term of (EP.11), including both endpoint terms, gives the exact overlapping product

$$\mathcal{K} := \prod_{j=0}^{r-1} (\epsilon_{n+j}^2 + \phi_{n+j}^2) = U \frac{V_n}{V_m} (1 - d_0)(1 - d_r) \prod_{j=1}^{r-1} (1 - d_j)^2. \quad (\text{EP.13})$$

Indeed the norm ratios telescope to U and $\prod_{j=1}^r d_j = V_m / V_n$. Each interior numerator occurs twice. As $E_*^{2r} \leq \prod \epsilon_{n+j}^2 \leq \mathcal{K}$ and $0 \leq 1 - d_j < 1$, this proves

$$E_* \leq (UV_n / V_m)^{1/(2r)}. \quad (\text{EP.14})$$

For $r \geq 2$ write $s = r - 1$ and retain all four endpoint ratios

$$\alpha = d_0 = V_n / V_{n-1}, \quad \beta = d_r = V_m / V_{m-1}, \quad t = \prod_{j=1}^s d_j = V_{m-1} / V_n. \quad (\text{EP.15})$$

For $0 < t < 1$, set $v_j = -\log d_j$. The function $f(v) = \log(1 - e^{-v})$ has $f''(v) = -e^{-v} / (1 - e^{-v})^2 < 0$ on $v > 0$. The tangent-line inequality $f(v) \leq f(\bar{v}) + f'(\bar{v})(v - \bar{v})$ follows by integrating the decreasing derivative on either side of $\bar{v}$. Summing at $\bar{v} = (-\log t) / s$ and exponentiating proves

$$\prod_{j=1}^s (1 - d_j) \leq (1 - t^{1/s})^s. \quad (\text{EP.16})$$

If one $d_j = 1$, the left side is zero; if $t = 1$, all are one and both sides are zero. For $t < 1$ with all $d_j < 1$, strict concavity shows equality exactly when all $d_j = t^{1/s}$. Insert this result in (EP.13) to obtain

$$E_* \leq \mathcal{U} := \mathcal{A} \left(\frac{(1 - \alpha)(1 - \beta)}{\beta t} \right)^{1/(2r)} (1 - t^{1/(r-1)})^{(r-1)/r}. \quad (\text{EP.17})$$

At $r = 1$ the separate, zero-safe formula is $E_* \leq \mathcal{A} \sqrt{(1 - \alpha)(\beta^{-1} - 1)}$. The four endpoint slots then overlap and no $t^{1/0}$ is defined.

This bound is attained in the relaxed scalar data class: choose the equal interior contractions and, when $\alpha, \beta, t < 1$, put $f_j = (1 - d_j)(d_{j+1}^{-1} - 1) > 0$ and

$$\omega_{n+j+1} / \omega_{n+j} = \mathcal{U}^2 / f_j, \quad \epsilon_{n+j} = \mathcal{U}, \quad \phi_{n+j} = 0. \quad (\text{EP.18})$$

Their product is U by (EP.17), so both norm endpoints are respected and every radius identity holds. If an endpoint contraction is one, or $t = 1$, every compatible sequence has a zero radius at some step, hence $E_* = 0$ and the zero bound is attained. At $r = 1$ choose the sole norm ratio U . These scalar data are related to the arithmetic objects by the evaluation map (EP.4)–(EP.12); the proof of scalar attainment does not assert that this evaluation map is surjective.

For strictly interior contractions define the exact loss

$$\mathcal{J} = s \log(1 - t^{1/s}) - \sum_{j=1}^s \log(1 - d_j) \geq 0. \quad \mathcal{K} = \mathcal{U}^{2r} e^{-2\mathcal{J}}. \quad (\text{EP.19})$$

It also has the proved quantitative lower bound

$$\mathcal{J} \geq \frac{t}{2(1-t)^2} \sum_{j=1}^s \left(v_j + \frac{\log t}{s} \right)^2. \quad (\text{EP.20})$$

To prove this, all v_j and their mean lie in $(0, -\log t]$, where $-f''(v) \geq t/(1-t)^2$. Integrating the derivative twice gives the tangent inequality with the displayed quadratic remainder. The linear terms sum to zero. This proves (EP.20), including $s = 1$, where both sides vanish. If an interior $d_j = 1$ the exact product (EP.13) applies with value zero; no logarithm of zero is evaluated.

Define the actual phase polynomial

$$P(y) = \prod_{j=0}^{r-1} (y + \phi_{n+j}^2), \quad y \geq 0. \quad P(E_*^2) \leq \mathcal{K}. \quad (\text{EP.21})$$

It is continuous, tends to infinity, and is strictly increasing on $[0, \infty)$: for $y_2 > y_1 \geq 0$ every factor increases, and if a factor vanishes at y_1 , the product at y_2 is strictly positive. The actual product guarantees $\mathcal{K} \geq P(0)$. There is therefore exactly one $y_* \geq 0$ with $P(y_*) = \mathcal{K}$, and $E_* \leq \sqrt{y_*}$. If $\mathcal{K} = P(0)$, then $y_* = 0$. If an externally supplied number were smaller than $P(0)$, it would be incompatible with the displayed radius data; it would not define a nonnegative threshold. In particular at $\mathcal{K} = 0$ at least one phase vanishes and $y_* = 0$. Positive rational enclosures for the phases' squares and $\mathcal{K}$ lead to polynomial evaluation and rational bisection; positive powers and products suffice for each comparison. Using $\mathcal{U}^{2r}$ in place of $\mathcal{K}$ gives the valid endpoint-only version, also with its phase retained.

3 The sharp scalar bound with only the combined budget

Put

$$\mathcal{B} = \log \frac{V_{n-1}V_n}{V_{m-1}V_m} = -\log \left(d_0 d_r \prod_{j=1}^{r-1} d_j^2 \right) \geq 0. \quad (\text{EP.22})$$

The product in (EP.13), divided by U , is the function

$$F(d) = \frac{(1-d_0)(1-d_r) \prod_{j=1}^{r-1} (1-d_j)^2}{d_r \prod_{j=1}^{r-1} d_j}. \quad (\text{EP.23})$$

For $\mathcal{B} > 0$ its unique maximum over these scalar contractions with this budget is attained at

$$d_0 = \frac{1}{1+2x}, \quad d_r = 1-2x, \quad d_j = \frac{1-x}{1+x} \quad (1 \leq j < r), \quad (\text{EP.24})$$

where $x \in (0, 1/2)$ is the unique solution of

$$\mathcal{B} = 2 \operatorname{artanh}(2x) + 4(r-1) \operatorname{artanh} x. \quad (\text{EP.25})$$

Here $\operatorname{artanh} x = \frac{1}{2} \log((1+x)/(1-x))$. The exact maximum and consequent bound are

$$F_{\max} = \frac{4^r x^{2r}}{(1-4x^2)(1-x^2)^{r-1}}, \quad \boxed{E_* \leq \mathcal{U} \leq \mathcal{A} \frac{2x}{[(1-4x^2)(1-x^2)^{r-1}]^{1/(2r)}}}. \quad (\text{EP.26})$$

The middle expression for $r = 1$ uses the separate expression after (EP.17).

Here is a full extremization proof. Write $v_j = -\log d_j$. The feasible simplex is $v_0 + v_r + 2 \sum_{j=1}^{r-1} v_j = \mathcal{B}$, $v_j \geq 0$, a compact set. The product vanishes at every boundary point and is positive at interior points, so a maximum is interior. Its logarithm is

$$f(v_0) + f(v_r) + 2 \sum_{j=1}^{r-1} f(v_j) + v_r + \sum_{j=1}^{r-1} v_j. \quad (\text{EP.27})$$

Its Hessian is strictly negative diagonal, so it is strictly concave. The Lagrange multiplier equations are

$$\frac{1}{e^{v_0} - 1} = \lambda, \quad 1 + \frac{1}{e^{v_r} - 1} = \lambda, \quad 1 + \frac{2}{e^{v_j} - 1} = 2\lambda \quad (1 \leq j < r). \quad (\text{EP.28})$$

They imply $\lambda > 1$ and yield (EP.24) with $x = 1/(2\lambda)$. The right side of (EP.25) is strictly increasing from zero to infinity because its derivative is $4/(1-4x^2) + 4(r-1)/(1-x^2) > 0$. Thus it has exactly one solution and determines the unique maximum. Substitution into (EP.23) gives (EP.26). The value $\mathcal{B} = 0$ forces all $d_j = 1$, and all bounds are zero; it is represented by the continuous endpoint $x = 0$. The norm-ratio construction (EP.18), now using these contractions and $\mathcal{A}F_{\max}^{1/(2r)}$, proves sharpness in the same scalar class for this reduced data.

We prove the comparison with the earlier hyperbolic-sine bound directly. For $a, b \in (0, 1]$ and $z = -\frac{1}{2} \log(ab)$,

$$\sinh^2 z - (1-a)(b^{-1}-1) = \frac{(1-2a+ab)^2}{4ab} \geq 0. \quad (\text{EP.29})$$

Expansion after multiplying by $4ab$ proves the equality term by term. Also $(\log \sinh z)'' = -1/\sinh^2 z < 0$ for $z > 0$, so the same tangent-line proof used for (EP.16) shows $\prod \sinh z_j \leq \sinh((\sum z_j)/r)^r$. If any $z_j = 0$, the product is zero and the inequality is immediate. Taking $a = d_j, b = d_{j+1}$ gives $\sum z_j = \mathcal{B}/2$, and hence

$$\mathcal{A}F_{\max}^{1/(2r)} \leq \mathcal{A} \sinh(\mathcal{B}/(2r)). \quad (\text{EP.30})$$

For $r > 1$, $\mathcal{B} > 0$ the inequality is strict. At a maximizer every $z_j > 0$. Equality in the concavity step requires all $z_j = z > 0$, so $d_j d_{j+1} = e^{-2z} =: p < 1$. Equality in (EP.29) requires $d_j = (1+p)/2$ for every $0 \leq j < r$. For $r > 1$ this implies $d_0 = d_1$, whence $p = ((1+p)/2)^2$ and $(1-p)^2 = 0$, a contradiction. For $r = 1$, (EP.25)–(EP.26) give exactly $\sinh(\mathcal{B}/2)$. For geometric contractions $d_j = e^{-v}$ the fixed-endpoint bound is $2\mathcal{A} \sinh(v/2)$ and the earlier bound is $\mathcal{A} \sinh v$; their ratio is exactly $\cosh(v/2)$.

The optimizer can be certified without a transcendental equation. The real endpoint datum $R = e^{\mathcal{B}} \geq 1$ gives

$$(1+2x)(1+x)^{2(r-1)} = R(1-2x)(1-x)^{2(r-1)}. \quad (\text{EP.31})$$

The ratio of the two positive sides before subtraction is strictly increasing on $[0, 1/2]$ by (EP.25), so rational bisection isolates its unique root using integer products when R is rational. For a general actual endpoint, certified rational bounds $1 \leq R_- \leq R \leq R_+$ give enclosing roots $x(R_-) \leq x(R) \leq x(R_+)$ by the same monotonicity. Positive-power bisection then encloses (EP.26). This specifies the computational map exactly.

4 Schur windows and the reduced source–relation determinant

Split the source monomials of $\mathcal{P}_m$ at degree $n-1$ and set

$$H_m = \begin{pmatrix} H_{n-1} & C_D \\ C_D^* & D_D \end{pmatrix}, \quad W = D_D - C_D^* H_{n-1}^{-1} C_D, \quad F = J_D - J_{n-1} H_{n-1}^{-1} C_D. \quad (\text{EP.32})$$

The triangular map $(x, y) \mapsto (x - H_{n-1}^{-1} C_D y, y)$, from $\mathbb{C}^n \oplus \mathbb{C}^{r+1}$ onto $\mathbb{C}^{m+1}$, has determinant one and takes H_m to $\text{diag}(H_{n-1}, W)$ and the remainder map to $[J_{n-1}, F]$. Consequently $W > 0$. For its first j columns put $W_j = W[1:j, 1:j]$, $F_j = F[:, 1:j]$, $Z_j = F_j^* G_{n-1} F_j$. Block inversion and the determinant identity from (EP.6) prove

$$K_{n+j-1} = K_{n-1} + F_j W_j^{-1} F_j^*, \quad R_j := V_{n-1}/V_{n+j-1} = \frac{\det(W_j + Z_j)}{\det W_j}, \quad R_0 = 1. \quad (\text{EP.33})$$

This R_j is a scalar and is distinct from the typed least-lift R_N by its explicit definition; the scalar is obtained from that least-lift Gram by determinant. Write

$$\begin{aligned} w &= (W_1)_{11}, \quad p = \det W_r, \quad s_+ = \det W_{r+1}, \quad e = w + (Z_1)_{11}, \\ H_+ &= \det(W_{r+1} + Z_{r+1}), \quad H_- = \det(W_r + Z_r). \end{aligned} \quad (\text{EP.34})$$

All six quantities are positive. Successive monic Schur complements give $w = \omega_n$ and $\omega_m = s_+/p$. Thus the exact bridge from the original EW matrices to the new endpoint quantities is

$$\begin{aligned} U &= s_+/(pw), \quad \alpha = w/e, \quad \beta = H_-s_+/(pH_+), \quad t = ep/(wH_-), \\ U \frac{V_n}{V_m} &= \frac{H_+}{pe}, \quad \mathcal{B} = \log \frac{H_-H_+w}{ps_+e}. \end{aligned} \quad (\text{EP.35})$$

For example $\alpha = 1/R_1$, $\beta = R_r/R_{r+1}$ and $t = R_1/R_r$, and substitution gives each displayed equality. Every individual contraction can also be recovered as $d_j = R_j/R_{j+1}$ for $0 \leq j \leq r$, so the exact product, the loss $\mathcal{J}$ and every comparison use the same block. For $r = 1$, $p = w$, $H_- = e$, $t = 1$, and the separate single-step formula applies. At $n = q$ the base relation space is empty and $G_{q-1} = H_{q-1}$; no inverse of an empty deficient kernel occurs.

To prove the determinant cancellation independently, define

$$\mathfrak{D}_j = \det H_{j-1}, \quad \mathfrak{D}_0 = 1, \quad \mathfrak{B}_l = \det(B_{q+l-1}^* H_{q+l-1} B_{q+l-1}), \quad \mathfrak{B}_0 = 1. \quad (\text{EP.36})$$

The coefficient matrix with first columns the q quotient monomials and last columns $\chi, S\chi, \dots, S^{N-q}\chi$ is triangular with diagonal one. Subtracting boundary columns to replace the first ones by R_N keeps its determinant one and makes the Gram block diagonal. Hence

$$\omega_N = \mathfrak{D}_{N+1}/\mathfrak{D}_N, \quad V_N = \mathfrak{D}_{N+1}/\mathfrak{B}_{N-q+1}, \quad \Lambda_N := \omega_N/V_N = \mathfrak{B}_{N-q+1}/\mathfrak{D}_N. \quad (\text{EP.37})$$

The norm identity uses the analogous monic orthogonal triangular basis. Equations (EP.14), (EP.35), and (EP.37) now prove

$$E_* \leq \left(\frac{\Lambda_m}{\Lambda_n} \right)^{1/(2r)} = \left(\frac{\mathfrak{B}_{m-q+1}\mathfrak{D}_n}{\mathfrak{D}_m\mathfrak{B}_{n-q+1}} \right)^{1/(2r)} = \left(\frac{H_+}{pe} \right)^{1/(2r)}. \quad (\text{EP.38})$$

For $N \geq q$, multiplication by χ in (EP.3) also sends the original leading-coefficient line at degree $N - q$ isomorphically to the leading-coefficient line of its relation at degree N . Its leading coefficient is one; the source norm of that relation is exactly the form defining $\mathfrak{B}$. The quotient J_N sends the same relation to zero. These give the maps relating the determinant and quotient observations.

For comparison only, multiplying the original source measure by a positive scalar a sends $H_N \mapsto aH_N$, $G_N \mapsto aG_N$, $\omega_N \mapsto a\omega_N$, $V_N \mapsto a^q V_N$, and $\Lambda_N \mapsto a^{1-q}\Lambda_N$. This follows directly from (EP.4) and (EP.37). All window ratios displayed above are unchanged; the actual measure throughout remains (EP.1).

5 Full confluent transfer and its two-determinant window

We now construct the transfer determinants in (EP.38) from their full raw derivative maps. Retain $\psi = \hat{\chi} = i^{-q}\chi(c + iu)$ and $\Pi = \psi^2$ under the reflection hypothesis of (EP.7). If χ has distinct roots λ_a of complete lengths ℓ_a , let $\zeta_a = (\lambda_a - c)/i$ and $r_a = 2\ell_a$. The width is $d = \sum_a r_a = 2q$. The map

$$\mathcal{J}_2 f = (f^{(e)}(\zeta_a))_{a, 0 \leq e < r_a} : \mathbb{C}[u] \longrightarrow \bigoplus_a \mathbb{C}^{r_a} \quad (\text{EP.38a})$$

uses the fixed order of nodes and then increasing raw derivative orders. Its kernel is (Π) by Taylor divisibility and the coprimality of distinct linear factors. In these coordinates a product uses $\sum_{b=0}^e \binom{e}{b} f^{(b)} g^{(e-b)}$, and multiplication by u has the block formula $(X_\Pi v)_{a,e} = \zeta_a v_{a,e} + e v_{a,e-1}$. The original coordinate map has the exact raw-jet expression

$$(\partial_u^e \Phi P)(\zeta_a) = i^e P^{(e)}(\lambda_a), \quad \mathbb{C}[S]/(\chi^2) \xrightarrow{\sim} \mathbb{C}[u]/(\Pi), \quad [P] \mapsto [P(c + iu)], \quad (\text{EP.38b})$$

since $\Phi(\chi^2) = i^{2q}\Pi$. Every $e!$ and i^e is retained.

Let $v_j = \mathcal{J}_2 Q_j$ for the same source polynomials used in (EP.8), and form

$$\mathbf{F}_a = [v_a, \dots, v_{a+d-1}], \quad \mathbf{V} = [\mathcal{J}_2 1, \dots, \mathcal{J}_2 u^{d-1}]. \quad (\text{EP.38c})$$

Their full raw Vandermonde determinant is

$$\det \mathbf{V} = \left(\prod_a \prod_{e=0}^{r_a-1} e! \right) \prod_{a < b} (\zeta_b - \zeta_a)^{r_a r_b} \neq 0. \quad (\text{EP.38d})$$

To prove the formula without changing node orders, replace the monomial columns successively by the monic polynomials $\prod_{b < a} (u - \zeta_b)^{r_b} (u - \zeta_a)^e$. This is a triangular column change of determinant one, ordered by total degree. Its jet matrix is block lower triangular. At its (a, e) diagonal entry, the derivative is $e! \prod_{b < a} (\zeta_a - \zeta_b)^{r_b}$. Multiplying these entries proves (EP.38d), including repeated orders. The matrix $\mathbf{F}_0$ equals $\mathbf{V}$ times the actual monic coefficient matrix; the determinants, rather than all entries, agree.

Every $\mathbf{F}_a$ is invertible. A vector in its kernel gives a polynomial $Q \in \text{span}(Q_a, \dots, Q_{a+d-1})$ with $Q = \Pi H$ and $\deg H < a$. Source orthogonality gives $0 = \langle H, Q \rangle = \int \Pi |H|^2 d\mu_\theta$. The polynomial Π is positive off its finitely many real roots, and the original source density is positive almost everywhere, so $H = 0$. For $a = 0$ divisibility and degree already force $Q = 0$. Solving $\mathbf{F}_a z_a = v_{a+d}$ yields a monic polynomial

$$\hat{Q}_a = \frac{Q_{a+d} - \sum_{j=0}^{d-1} (z_a)_j Q_{a+j}}{\Pi}. \quad (\text{EP.38e})$$

It has degree a and is orthogonal for the weight $\Pi d\mu_\theta$: pair its numerator with any polynomial of smaller degree. The unique monic orthogonal polynomial for this real positive weight is real, so this quotient is real. Pairing its numerator with itself after division, or with $\hat{Q}_a$, leaves only the Q_a term and gives

$$\nu_a := \int \Pi |\hat{Q}_a|^2 d\mu_\theta = -(z_a)_0 \omega_a > 0. \quad (\text{EP.38f})$$

All source monic polynomials have been left at their literal norms. The original monic relation polynomial is $\chi(S) i^a \hat{Q}_a((S - c)/i)$, whose value on the source line is $i^{a+q} \psi(u) \hat{Q}_a(u)$ and whose norm is exactly ν_a . This is the explicit source map to the relation norm in (EP.36).

The column shift gives $\mathbf{F}_{a+1} = \mathbf{F}_a[e_1, \dots, e_{d-1}, z_a]$. Its determinant multiplier is $(-1)^{d-1} (z_a)_0 = -(z_a)_0$ because $d = 2q$. Multiplying (EP.38f) from index zero proves

$$\boxed{\frac{\det \mathbf{F}_a}{\det \mathbf{V}} = \prod_{j=0}^{a-1} \frac{\nu_j}{\omega_j} = \frac{\mathfrak{B}_a}{\mathfrak{D}_a}}. \quad (\text{EP.38g})$$

The source and relation determinant identities follow from their monic orthogonal triangular matrices; the S - u coefficient matrices have determinant of modulus one, including the common i^q on each relation column. Equation (EP.38g) therefore holds for the original determinants, with their complex node factors intact.

For $a = N - q + 1$, substitution into (EP.37) gives

$$V_N = \left(\prod_{j=a}^N \omega_j \right) \frac{\det \mathbf{V}}{\det \mathbf{F}_a}, \quad \Lambda_N = \frac{\det \mathbf{F}_a}{\det \mathbf{V} \prod_{j=a}^{N-1} \omega_j}. \quad (\text{EP.38h})$$

In particular with $a = n - q + 1$ the two-volume norm criterion is

$$\boxed{E_* \leq \left[\frac{\det \mathbf{F}_{a+r}}{\det \mathbf{F}_a} \prod_{j=0}^{q-2} \frac{\omega_{a+j}}{\omega_{a+r+j}} \right]^{1/(2r)}}. \quad (\text{EP.38i})$$

For $q = 1$ the product has zero terms and value one. The expression inside the root is positive by its exact equality to Λ_m/Λ_n ; no absolute value conceals a complex phase.

The complete transfer of the jet multiplication map is also explicit. Put $K_a = [e_1, \dots, e_{d-1}, z_a]$ and $T_a = F_a^{-1} X_\Pi F_a$. Then $T_{a+1} = K_a^{-1} T_a K_a$, and the source recurrence gives $z_a = (T_a - a_{a+d-1}^{(0)} I) e_{d-1} - (\omega_{a+d-1}/\omega_{a+d-2}) e_{d-2}$. Here $d \geq 2$, so both indices exist. Differentiating monic orthogonality gives $\partial_\theta Q_j = -(\omega_j/\omega_{j-1}) Q_{j-1}$ for $j \geq 1$ and $\partial_\theta Q_0 = 0$: its pairings below $j-1$ vanish and its Q_{j-1} pairing is $-\omega_j$. The norm derivative is $\partial_\theta \log \omega_j = a_j^{(0)}$ by the same minimum argument as (EP.9). Internal shifted columns have zero trace, so

$$\partial_\theta \log V_N = \sum_{j=a}^N a_j^{(0)} + \frac{\omega_a}{\omega_{a-1}} e_0^T F_a^{-1} v_{a-1}. \quad (\text{EP.38j})$$

At $a = 0$ omit the last term entirely. This follows by writing $F'_a = F_a L_a - (\omega_a/\omega_{a-1}) v_{a-1} e_0^T$, where $(L_a)_{j-1,j} = -\omega_{a+j}/\omega_{a+j-1}$ and all other entries vanish. Taking the trace of $F_a^{-1} F'_a$ in (EP.38h) proves (EP.38j). It supplies the phase in (EP.9) through the same complete jet map.

For the Gamma reference use the same Π , nodes, row order, and raw derivatives, and denote the reference by a superscript Γ . With $X_a = \mathfrak{D}_a/\mathfrak{D}_a^\Gamma$ and $Y_a = \mathfrak{B}_a/\mathfrak{B}_a^\Gamma$, division of (EP.38g) gives

$$\frac{\det F_a}{\det F_a^\Gamma} = \frac{Y_a}{X_a}, \quad \frac{V_N}{V_N^\Gamma} = \frac{X_{N+1}}{Y_{N-q+1}}. \quad (\text{EP.38k})$$

The invariant endpoint values still depend only on original moments through degree $2m$ by (EP.32)–(EP.38). In contrast the individual entries of F_a , with $a = N-q+1$, involve Q_{N+q} , which is obtained by a projection using moments through $2N+2q-1$. The determinant identity proves their exact cancellation in the invariant answer. It does not identify every raw entry with a function of the shorter input list.

For a non-reflection-stable polynomial, define $\chi^\dagger(S) = (-1)^q \bar{\chi}(k-S)$, where bars conjugate coefficients. Then $\Phi\chi = i^q \psi$ and $\Phi\chi^\dagger = i^q \bar{\psi}$, so the exact extension of the norm-jet map has domain $\mathbb{C}[S]/(\chi\chi^\dagger) \xrightarrow{\sim} \mathbb{C}[u]/(\bar{\psi}\psi)$ under the same Φ . It uses $\Pi = \bar{\psi}\psi$ instead of ψ^2 . Its nodes are the union of the roots of ψ and their conjugates, and the orders add at every overlap; their sum is again $2q$. The positivity, division, raw-jet, and determinant proofs above apply to this actual monic Π , with the just-proved changed quotient domain. The original relation map still multiplies by χ , whose squared norm weight is $|\psi|^2$. This supplies the exact general norm map. The original radius and phase argument (EP.7)–(EP.11) is stated under its reflection hypothesis and is not extended by this norm observation alone.

6 An arithmetic lower cost on one relation block

The complete analytic theorem in the delivered arithmetic endpoint source proves, for the actual density (EP.1) at $\theta = 0$,

$$(\omega_{h,k,2n}/\omega_{h,k,n})^{1/(2n)} \leq C_h n \quad (n \geq k \geq 3). \quad (\text{EP.39})$$

Its retained literal sufficient constant, from equation (36) there, is

$$C_h = 64b^{-2} e^{\pi/2} 2^{B_h/2} e^{B_h/(2e)} \sqrt{M_* \max(1, c_h^{-1}) \max(1, \vartheta_h^{-1})}, \quad 0 < b < \pi/2, \quad (\text{EP.40})$$

where $B_h = 42 + 2 \deg h$, $M_* = \max(1, M_h(b), M_h(-b))$, $M_h(b) = \int e^{bt} w_h(t) dt$, $\vartheta_h = \int_{-1}^1 w_h(t) dt$, and $c_h = b_h c_h^{(1)} > 0$ with $b_h = \min_{|u| \leq 1} (w_h * w_h)(u)$ and the original local-mass constant $c_h^{(1)}$ of that source. No norm estimate is assumed in this paper in place of that proof. Its complete source is retained in the dependency record, with the independent analytic audit.

For a packet consisting exactly of an off-line quartet $1/2 \pm \delta \pm i\gamma$, $\delta, \gamma > 0$, with common full multiplicity m_0 , the original exterior calculation gives

$$\begin{aligned} q_k &= [1 + k(m_0 - 1)](k + 1)^2, \\ L_{h,k} &= 2\delta[1 + k(m_0 - 1)](k + 1) \left\lfloor \frac{(k + 1)^2}{4} \right\rfloor, \\ L_{h,k}/q_k &\geq \delta k/2, \quad L_{h,k} \leq \epsilon_{h,k,N}. \end{aligned} \tag{EP.41}$$

Here is the complete finite spectral calculation behind (EP.41). In each tensor local algebra write $s_j = \rho_j + z_j$ with $z_j^{m_0} = 0$. The sum nilpotent $\sum z_j$ has exact order $1 + k(m_0 - 1)$: its top nonzero power has coefficient $[k(m_0 - 1)]! / [(m_0 - 1)!]^k$ at $\prod z_j^{m_0 - 1}$, and every higher monomial has an exponent at least m_0 . The distinct sum centres are $c + \delta(2a - k) + i\gamma(2b - k)$, $0 \leq a, b \leq k$. Every such pair occurs: the number of simultaneous plus signs can be any integer between $\max(0, a + b - k)$ and $\min(a, b)$, a nonempty interval. Distinct pairs give distinct centres since $\delta, \gamma > 0$. The cyclic annihilator is the least common multiple of these local annihilators, giving the displayed q_k with the full common order.

For any admitted G_N , conjugate $A - cI$ to the Euclidean metric by $G_N^{1/2}$ and put it in unitary upper triangular form, ordering the eigenvalues with positive real part first. Such a form follows inductively by choosing an eigenvector and applying induction to the quotient orthogonal block. Its Hermitian sum has eigenvalues $\epsilon_N, -\epsilon_N$ and zeros by (EP.10)–(EP.11). The trace of its compression onto the positive-real diagonal slots is twice the sum of those positive real parts. This trace is at most ϵ_N : in an eigenbasis of the Hermitian sum it equals $\epsilon_N \langle Pe_+, e_+ \rangle - \epsilon_N \langle Pe_-, e_- \rangle \leq \epsilon_N$ for the orthogonal projection P onto those slots. The sum of the positive integers $2a - k$, $0 \leq a \leq k$, is $\lfloor (k + 1)^2/4 \rfloor$, by separately summing the even and odd arithmetic progressions. Multiplying by 2δ , the $k + 1$ imaginary centres, and the full local length proves exactly $L_{h,k}$ and its bound by ϵ_N . Finally $\lfloor (k + 1)^2/4 \rfloor \geq k(k + 1)/4$, checked in the same two parities, proves $L_{h,k}/q_k \geq \delta k/2$. Thus the last inequality in (EP.41) holds at every admitted degree. Retaining the exact $L_{h,k}$ makes the following deduction explicit. Set $n = r = q_k$ in (EP.13), and abbreviate $q = q_k$, $L = L_{h,k}$. Every $d_j < 1$ in this window because $L > 0$ and (EP.11). Taking logarithms in (EP.13) and using (EP.39) proves

$$\begin{aligned} \log \frac{V_q}{V_{2q}} &\geq 2q \log \frac{L}{C_h q} - \log(1 - d_0) - \log(1 - d_q) - 2 \sum_{j=1}^{q-1} \log(1 - d_j) \\ &\quad + \sum_{j=0}^{q-1} \log(1 + \phi_{q+j}^2/L^2). \end{aligned} \tag{EP.42}$$

Indeed $\prod(\epsilon^2 + \phi^2) \geq L^{2q} \prod(1 + \phi^2/L^2)$, whereas $\log U \leq 2q \log(C_h q)$. All added terms on the right of (EP.42) after its first one are nonnegative, with the contraction penalty strictly positive. For $q > 1$, (EP.16) gives the additional endpoint-only lower penalty $-\log(1 - \alpha) - \log(1 - \beta) - 2(q - 1) \log(1 - t^{1/(q-1)})$. In particular the actual two-volume threshold is

$$\boxed{\liminf_{k \rightarrow \infty} \frac{\log(V_{q_k}/V_{2q_k})}{q_k \log k} \geq 2.} \tag{EP.43}$$

This follows by replacing L/q_k by $\delta k/2$ in (EP.42) and dividing by $q_k \log k$. The constants are independent of k . The first norm window also gives the exact determinant lower bound $\Lambda_{2q}/\Lambda_q \geq L^{2q}$ by (EP.38).

Here is the exact receiving relation layer. For $i \geq q - 1$, $j > i$, let $\mathbf{T} = [p_{i+1}, \dots, p_j]$ be source columns, $\mathbf{F} = [b_{i+1}, \dots, b_j]$, and $\Omega = \text{diag}(\omega_{i+1}, \dots, \omega_j)$. Then

$$\mathbf{b} = \mathbf{T} - R_i \mathbf{F} : \mathbb{C}^{j-i} \longrightarrow \mathcal{D}_j \cap \mathcal{D}_i^\perp, \quad \mathcal{D}_N = \chi \mathcal{P}_{N-q}, \quad \mathbf{b}^* \mathbf{b} = \Omega + \mathbf{F}^* G_i^* \mathbf{F}. \tag{EP.44}$$

Reduction vanishes termwise. Each column is orthogonal to $\mathcal{D}_i$ because $p_{i+a} \perp \mathcal{P}_i$ and $R_i \perp \mathcal{D}_i$. Their highest degrees are distinct and monic. Successive subtraction of these highest coefficients spans the dimension- $j-i$ new relation layer and proves that $\mathfrak{b}$ is an isomorphism onto that layer. Orthogonality of the two summands gives its exact Gram. Kernel addition as in (EP.5) yields

$$\frac{V_i}{V_j} = \det(I + \Omega^{-1/2} \mathbf{F}^* G_i \mathbf{F} \Omega^{-1/2}) = \prod_{a=1}^{s_0} (1 + \lambda_a), \quad s_0 = \text{rank } \mathbf{F} \leq \min(q, j-i). \quad (\text{EP.45})$$

The positive eigenvalues λ_a are those of the written positive semidefinite matrix; zero eigenvalues contribute one. At $(i, j) = (q, 2q)$, (EP.42) thus gives

$$\sum_a \log(1 + \lambda_a) \geq 2q \log \frac{L}{C_h q}, \quad \lambda_{\max} \geq \left(\frac{L}{C_h q} \right)^2 - 1 \geq \frac{\delta^2 k^2}{4C_h^2} - 1. \quad (\text{EP.46})$$

For the second assertion, pad the eigenvalue list to length q with zeros, define $\lambda_{\max}$ as the maximum of this full list, and bound the determinant by $(1 + \lambda_{\max})^q$. If its right side is negative, the assertion still follows from $\lambda_{\max} \geq 0$. For sufficiently large k the determinant lower bound exceeds one, so the layer has positive rank.

The representative change is determined, not discarded: projection onto the new orthogonal relation layer in (EP.44) gives

$$R_j = R_i + \mathfrak{b}(\Omega + \mathbf{F}^* G_i \mathbf{F})^{-1} \mathbf{F}^* G_i. \quad (\text{EP.47})$$

Indeed $\mathfrak{b}^* R_i = -\mathbf{F}^* G_i$, so the right side is a lift orthogonal to both old and new relations, hence the unique minimum lift. Its arithmetic jet remains $\eta x = [v_h]_h^{\otimes k} x(S)$, whose injectivity follows because the full local unit has a recursive power-series inverse and (χ) is exactly the sum-polynomial kernel. For an actual relation $\chi\kappa$, repeated monic division gives polynomials Q_a with $\chi(\sum s_a) = \sum_a h(s_a) Q_a(\mathbf{s})$. The original cochain

$$\sum_{a=1}^k (-1)^{a-1} Q_a(\mathbf{D}) \kappa \left(\sum D_b \right) (F_h^{\otimes(a-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-a)}) \quad (\text{EP.48})$$

has differential $\mathcal{V}_{h,k}(\chi\kappa)$, since $h(D)F_h = \Theta\phi_0$ and the tensor differential contributes the same sign $(-1)^{a-1}$, with all other differentials zero. Thus its quotient image is the supported zero $(\ell, 0)$ at the same label ℓ . The fibre map sends (ℓ, x) to $(\ell, R_N x)$ and the external absence τ to τ . This identifies the retained source cost in (EP.44)–(EP.46) with the relation sent to that supported zero through the actual maps.

The four-volume budget is linked exactly to the two-volume cost by

$$\mathcal{B} = 2 \log(V_n/V_m) + \log(\beta/\alpha). \quad (\text{EP.49})$$

The boundary term remains signed. The two-volume threshold alone does not bound it from below. A different, balanced window connects the original four endpoints without that signed comparison: taking $n = q, r = q - 1$ yields the central cost $\log(V_q/V_{2q-1})$, and monotonicity proves

$$\log \frac{V_{q-1} V_q}{V_{2q-1} V_{2q}} \geq 2 \log \frac{V_q}{V_{2q-1}}. \quad (\text{EP.50})$$

The complete proof in the accompanying *Four-volume threshold* continuation, read in full, establishes

$$(\omega_{h,k,n+r}/\omega_{h,k,n})^{1/(2r)} \leq C_h^{\text{bal}} n, \quad n \geq k \geq 3, \quad n/2 \leq r \leq n, \quad (\text{EP.50a})$$

with the literal sufficient constant

$$C_h^{\text{bal}} = 256e^\pi \max(1, 3/c_h) M_* \max(1, \vartheta_h^{-1}) \max(1, b^{-3}) 6^{B_h/3}. \quad (\text{EP.50b})$$

It derives this from the original lower convolution envelope and the exact monic Legendre norm, retaining every factor before its $2r$ -th root. Its full source proof is included as a dependency. Applying (EP.13) now at $n = q, r = q - 1$ gives the following complete version with its retained penalties:

$$C_{\text{mid}} := \log \frac{V_q}{V_{2q-1}} \geq 2(q-1) \log \frac{L}{C_h^{\text{bal}} q} - \log(1 - d_0) - \log(1 - d_{q-1}) - 2 \sum_{j=1}^{q-2} \log(1 - d_j) + \sum_{j=0}^{q-2} \log(1 + \phi_{q+j}^2/L^2), \quad d_j = \delta_{q+j}. \quad (\text{EP.50c})$$

All $d_j < 1$ follow again from $L > 0$ in (EP.11). The original four-volume budget has the exact positive decomposition

$$\log \frac{V_{q-1} V_q}{V_{2q-1} V_{2q}} = 2C_{\text{mid}} + \log \frac{V_{q-1}}{V_q} + \log \frac{V_{2q-1}}{V_{2q}}. \quad (\text{EP.50d})$$

Both extra terms are nonnegative. Dropping only nonnegative terms from (EP.50c)–(EP.50d) and using $L/q \geq \delta k/2$ proves

$$\liminf_{k \rightarrow \infty} \frac{\log(V_{q_k-1} V_{q_k} / (V_{2q_k-1} V_{2q_k}))}{q_k \log k} \geq 4. \quad (\text{EP.50e})$$

Indeed the numerator is at least $4(q_k - 1)[\log k + \log(\delta/(2C_h^{\text{bal}}))]$ and $(q_k - 1)/q_k \rightarrow 1$. For the actual central relation block $(i, j) = (q, 2q - 1)$, its full matrix in (EP.45) has size $q - 1$, so the same determinant argument gives

$$\lambda_{\text{max}}^{\text{mid}} \geq \left(L / (C_h^{\text{bal}} q) \right)^2 - 1 \geq \delta^2 k^2 / (4(C_h^{\text{bal}})^2) - 1. \quad (\text{EP.50f})$$

The norm bound, the new product penalties, the central relation eigenvalues, and the original four-volume budget are therefore connected by these explicit maps. No upper estimate for those growing arithmetic relation matrices is proved by this lower-cost calculation.

7 Mass-seven polynomial calibration and zero cases

The independent finite fixture is the measure $7e^{-u^2/2} du / \sqrt{2\pi}$ and quotient $\mathbb{C}[u]/(u^2 + 1)$. Integration by parts gives monic Hermite polynomials $Q_0 = 1, Q_1 = u, Q_{j+1} = uQ_j - jQ_{j-1}$ and norms $\omega_j = 7j!$: the derivative identity $Q'_j = jQ_{j-1}$ follows by induction in the recurrence, and integration by parts proves orthogonality and $\|Q_j\|^2 = j\|Q_{j-1}\|^2$ starting from mass seven. In the quotient basis $(1, u)$ use

$$T = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad A = \frac{1}{2}I + iT. \quad (\text{EP.51})$$

Its original- S relation is $\chi(S) = (S - 1/2)^2 - 1$. The exact map from the $(1, S)$ coefficients to $(1, u)$ is $E = \begin{pmatrix} 1 & 1/2 \\ 0 & i \end{pmatrix}$; $G_N^S = E^* G_N^u E$ and $|\det E| = 1$. Thus all fixture volumes are the original- S volumes as well. Kernel summation (EP.5), in the transported coordinates, gives

$$G_2^u = \text{diag}(7/3, 7), \quad G_3^u = \text{diag}(7/3, 21/11), \quad G_4^u = \text{diag}(42/43, 21/11), \quad (V_1, V_2, V_3, V_4, V_5) = (49, 49/3, 49/11, 882/473, 980/1333). \quad (\text{EP.52})$$

For a diagonal $G = \text{diag}(g_0, g_1)$, $A^*G + GA - G = i(GT - T^T G)$ has relative squared radius $(g_0 + g_1)^2 / (g_0 g_1)$. The resulting exact values are

$$(\epsilon_2^2, \epsilon_3^2, \epsilon_4^2) = (16/3, 400/99, 4225/946), \quad U_{2,3} = 60, \quad (\alpha, \beta, t) = (1/3, 110/279, 54/473). \quad (\text{EP.53})$$

Reflection $u \mapsto -u$ conjugates the source and quotient matrices at tilt θ to those at $-\theta$ by diagonal sign matrices. Their determinants are even, and $\text{Tr } T = 0$, so $\phi_N = 0$ at zero tilt for this fixture. This proves that the direct defect and the radius identities test the same allowance.

All finite values in (EP.52)–(EP.53) are derived in the checker from literal Gaussian moments $7(2j-1)!!$, monomial Gram matrices, monic polynomial division, and (EP.4), independently of assigned quotient matrices. Further windows check (EP.13), (EP.35), and (EP.38), while exact scalar extremizers check the new formula and phase boundary cases. These are finite auxiliary calculations. They do not assert that the Gaussian fixture is the arithmetic convolution measure; (EP.1)–(EP.7), (EP.32)–(EP.38) give the explicit common maps and the distinct specified source inputs.

For the mass-seven window, the exact inputs to the new reduced-budget root calculation are $U = 60$ and $e^B = 630509/1080$. The rational procedure (EP.31), followed by the positive sixth-root comparisons in (EP.26), gives the explicit certificate

$$\frac{719736229856370310835738215}{2^{88}} \leq \mathcal{A}F_{\max}^{1/6} \leq \frac{719736229856370310835738217}{2^{88}}. \quad (\text{EP.54})$$

Its value is approximately 2.325593185504684, with that decimal used only for orientation. The check record contains the two rational brackets for x , their exact comparisons in (EP.31), and the two sixth-power comparisons determining (EP.54).

The primary checker executes 415 exact auxiliary mathematical checks normally and under `python -O`. Its six actual formula changes alter the literal mass, the repeated interior contraction, the terminal ratio orientation, the Schur cancellation, the retained phase, and a raw derivative factorial; each changed run executes the full suite and is rejected in both modes. An independent scalar checker executes 2,453 exact checks in its unmodified runs, and its five changed-formula runs are rejected in both modes. Repeated modes and changed-formula runs do not add mathematical checks to those scopes. Their full source, complete records, independent mathematical reviews, and the eight exact proof dependencies accompany this article.

For the empty quotient, declare $q = 0$, $\chi = 1$, $C = 0$ and $V_N = 1$ including $N = -1$. Then $\epsilon_N = \phi_N = 0$, all quotient maps and relation-kernel update columns have their empty dimensions, and the exact product is zero. The formulas requiring $q \geq 1$ are used only in their stated range; the cochain boundary map and original source mass still exist. For $q = 1$ the transfer norm bands, when used, are empty products equal to one. At an endpoint contraction equal to one all zero-radius and phase cases are handled by (EP.11), (EP.13), and (EP.21) before logarithms.